Type MS · Use image size, actual size and magnification

Answer 1 of 13 · Question 3 part b.i · 2 marks

2023 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

3 (a)	• contains particular features / characteristics/ adaptations; • to perform a specific function;		2
(b) (i)	• 84 ÷ 0.006; • 14000x	Allow correct final answer from a calculation that includes 84 and 0.006, max 1 mark	2
(ii)	half the amount/haploid/23 chromosomes;		1
(iii)	• to provide energy; • for swimming/movement/travelling;		2
(iv)	• 180 ÷ 0.070; • 2571; • (÷ 60) 42.85;	Allow any figure that rounds to 43 Allow ecf for third mark - an incorrect value for a calculation carried out for mps1 and 2, divided by 60 **and** a correct answer to that calculation	3 exp
(c)	Any two from: • greater magnification; • higher resolution/can see objects more clearly/in more detail; • can see a greater number of structures;		2 exp

Total Question 3 = 12 marks

Type MS · Use image size, actual size and magnification

Answer 2 of 13 · Question 2 part a.iii · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

2 (a)	(i)	<table><tr><th>Part</th><th>Name of part</th><th>Function</th></tr><tr><td>X</td><td>mitochondria;</td><td></td></tr><tr><td>Y</td><td></td><td>controls activities of the cell/stores DNA / genetic information;</td></tr></table>		1 1
	(ii)	**A,** more cell structures can be seen with greater resolution		1
	(iii)	60mm/6cm; 60 ÷ 0.05; 1200;	Ecf two marks for correct calculation from incorrect measurement Full marks for correct final answer	3
	(b)	• diffusion; • from a high concentration (in the cell) to a lower concentration (in the blood)/down a concentration gradient;		1 1

Total question 2 = 8 marks

Type MS · Use image size, actual size and magnification

Answer 2 of 13 · Question 2 part a.iii · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use image size, actual size and magnification

Answer 3 of 13 · Question 5 part a.i · 2 marks

2020 JAN · Paper 01 · find part a.i in this mark scheme extract

number				
5 (a)	(i)	12 x 50 = 600 (1) 600 ÷ 1000 = 0.6 (1)	Allow 600 without working out. Allow full marks for correct final answer.	2
	(ii)	provides insulation/prevents nerve impulse dispersing/speeds up nerve impulse (1)		1
(b)	(i)	• hormones are transported in blood (1) • speed of blood flow is determined by heart rate/hormones can only travel at the speed blood is moving/blood flow slower (1)	Allow reverse argument for both marking points	2
	(ii)	• recognition of distance divided by time (1) • $1.0 \div 1.2 \times 10^{2}$ / $1.0 \times 10^{6} \div 1.2 \times 10^{8}$ (1) • 0.0083(3) / 8.3×10^{-3} (s) (1)	Allow full marks for correct final answer	3

Total for Question 5 = 8 marks

Type MS · Use image size, actual size and magnification

Answer 4 of 13 · Question 3 part b.i · 1 mark

2020 NOV · Paper 01 (regional R) · find part b.i in this mark scheme extract

3	(a) (i)	<table><tr><th></th><th>Steps</th></tr><tr><td>4</td><td>select a suitable objective lens</td></tr><tr><td>5</td><td>adjust focussing wheel to obtain a clear image</td></tr><tr><td>1</td><td>remove cells from the inside of the cheek</td></tr><tr><td>3</td><td>place microscope slide onto the microscope stage</td></tr><tr><td>2</td><td>smear the cell sample onto a microscope slide</td></tr></table>	One mark for 4 before 5. One mark for 3 before 2	2
	(ii)	more detail/organelles can be seen/organelles/named organelle seen more clearly (1)		1
	(iii)	reference no stain applied to specimen(1)		1
	(iv)	• reference to sterilising (named) equipment (1) • wash hands to avoid cross contamination(1) • incubate/store in sealed petri dish / container	allow valid sterilisation technique	2
	(b) (i)	10 × 40 = 400	Full marks for correct answer without working	1
	(ii)	reference to 1000 (1) 0.018 × 1000 = 18 (µm)	Full marks for correct final answer	2

Total Question 3 = 9 marks

Type MS · Use image size, actual size and magnification

Answer 5 of 13 · Question 5 part c · 4 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

number				
5 (a)	(i)	• join with ovum; • during fertilisation;		2
	(ii)	any two from • ribosomes; • endoplasmic reticulum/ER; • chromosomes;		2
(b)		• site of (aerobic) respiration; • energy released/ATP produced; • sperm motile/active/swim;		3
(c)		3µm = 0.003 mm; distance between X and Y = 145 mm; 145; 0.003 =48333/48000;	ECF = 2	4

Total 11 marks

Type MS · Use image size, actual size and magnification

Answer 5 of 13 · Question 5 part c · 4 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use image size, actual size and magnification

Answer 6 of 13 · Question 7 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
7 (a)	(i)	• prophase, metaphase, anaphase and telophase;;	minus 1 for each error	2 marks
	(ii)	• chromosomes lined up; • equator shown/aligned at equator; • one member of each bivalent above, the other below;		3 marks
(b)	(i)	A (DNA); B not part of DNA C not part of DNA D found only in RNA not DNA		1 mark
	(ii)	length = 20 mm; $\frac{20}{750}$ = 0.026 mm; = 26.0/26.7 µm;	allow 20 -22 ECF	3 marks
(c)		• two genetically identical cells; • diploid/2n;		2 marks

Total 11 marks

Type MS · Use image size, actual size and magnification

Answer 6 of 13 · Question 7 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use image size, actual size and magnification

Answer 7 of 13 · Question 1 part b.iii · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks
1 (a) (i)	• transport oxygen; • from lungs; • to (body) tissues/cells;	Allow organs	3
(ii)	• haemoglobin to carry/combine with oxygen; • no nucleus, so more haemoglobin; • biconcave shape to increase surface area; • small/flexible (cell) membrane to pass through capillaries; • thin membrane to reduce diffusion distance;		4
(b) (i)	vein;		1
(ii)	• valve present/large lumen/thin (muscular) walls; • only found in veins/not found in arteries/other blood vessels;		2
(iii)	• distance 13-15 mm; • divide by 10 = 1.3-1.5;	Ecf: max 1 mark if mp1 incorrect but division by 10 of their value is correct	2 Total 12

Question number	Answer	Notes	Marks

Type MS · Use image size, actual size and magnification

Answer 7 of 13 · Question 1 part b.iii · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Type MS · Use image size, actual size and magnification

Answer 8 of 13 · Question 1 part a.ii · 3 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number					
1	(a)	(i)	• first convoluted tubule; • loop of Henlé; • second convoluted tubule;	diagram only, no labels required	3
		(ii)	44mm; 44 ÷ 650; = 0.068mm;	ecf from mp1 max 2 marks Allow 0.0676	3
		(iii)	• protein; • red/white blood cells/platelets;		2
		(iv)	glucose;		1
	(b)	(i)	Any 2 from: • same person; • same activity levels; • same volume of water; • same food eaten;		2
		(ii)	710 ÷ 4; = 178 cm^3 per hour;	Allow 1 mark if division by 4 is shown Allow 177.5	2
		(iii)	• less urine production/lower urine loss on day two; • more ADH secreted on second day; • more water reabsorbed; • more water lost by sweating;	Allow reverse argument for all marking points	3

Total 16 marks

Type MS · Use image size, actual size and magnification

Answer 8 of 13 · Question 1 part a.ii · 3 marks

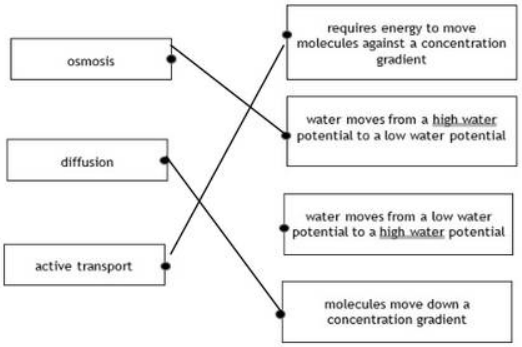
2022 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use image size, actual size and magnification

Answer 9 of 13 · Question 1 part c.ii · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c.ii in this mark scheme extract

Number				
1	(a) (i)	X = cell membrane; Y = nucleus; Z = cytoplasm;	Allow nuclear membrane	3
	(ii)	mitochondria/ribosomes/endoplasmic reticulum		1
	(iii)	• similar cells work together to form tissues; • (different) tissues work together to form organs;		2
	(b)		Reject more than one line from each process	3
	(c) (i)	A = squamous (epithelium) B = ciliated (epithelium)		2
	(ii)	60 x 40; 2400 (µm) (÷1000); 2.4 (mm);	Ecf from mp1. Max 2 marks e.g. 60 ÷ 40 (÷1000) = 0.0015	3

Total 14 marks

Type MS · Use image size, actual size and magnification

Answer 9 of 13 · Question 1 part c.ii · 3 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use image size, actual size and magnification

Answer 10 of 13 · Question 4 part a.ii · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
4	(a)	(i)	cell wall/capsule/slime coat;		1
		(ii)	55 ÷ 30 000 = 0.0018: 1.8/1.83/1.833;		2
	(b)	(i)	more densely/higher populated/more people/overcrowded/more waste/rubbish/sewage;		1
		(ii)	• fly lands on faeces (of person with a stomach ulcer); • transfers the bacteria to food; • person ingests the contaminated food;		3
	(c)	(i)	Two from: • high/medium levels in less well developed/poorer countries; • where sewage treatment is less effective/water not clean/poor sanitation; • more densely/highly populated; • healthcare/vaccines to treat infected people is less effective/available;	accept reverse arguments	2
		(ii)	Five from: • bacteria mutate/acquire resistant plasmids; • people do not finish the course/over-use of antibiotics; • resistant bacteria survive (and non-resistant bacteria die); • (resistant bacteria) reproduce/multiply; • offspring are antibiotic resistant/not affected by antibiotics; • antibiotic that will be effective is needed/used; • more serious infection occurs;		5

Total 14 marks

Type MS · Use image size, actual size and magnification

Answer 10 of 13 · Question 4 part a.ii · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use image size, actual size and magnification

Answer 11 of 13 · Question 6 part a.iii · 4 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number				
6	(a) (i)	A = mitochondria; B = nucleus; C = acrosome;		3
	(ii)	Any three from • releases enzymes; • digestion of outer part of ovum/jelly coat; • to allow nucleus of sperm to enter; • reference to fertilisation;	Ignore contains / produces enzymes Ignore cell membrane Ignore head	3
	(iii)	• length on diagram is 122 (mm); • = 122 000(µm); • 122 000 ; • 50 • = 2440;	Allow a value between 120 and 122	4
	(b)	• release energy/produce ATP; • by (aerobic) respiration; • allows swimming/movement of sperm (to ovum);		3

Total 13 marks

Type MS · Use image size, actual size and magnification

Answer 11 of 13 · Question 6 part a.iii · 4 marks

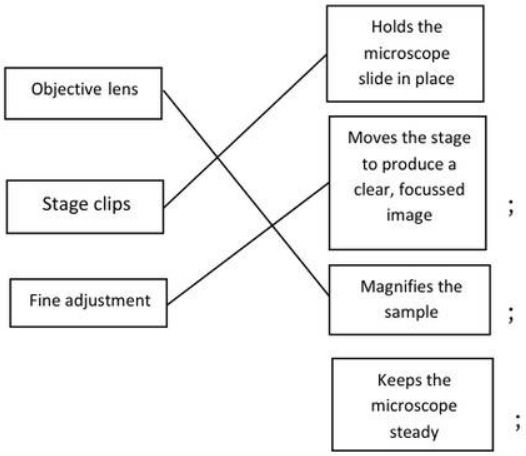
2023 NOV · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use image size, actual size and magnification

Answer 12 of 13 · Question 1 part b.ii · 3 marks

2024 JUN · Paper 01 (regional R) · find part b.ii in this mark scheme extract

Number				
1	(a) (i)	**Any two from:** • (both) have a cell membrane; • (both) have cytoplasm; • (both) have genetic material/DNA; • (both) contain ribosomes;		(2)
	(a) (ii)	**Any two from:** • bacterial cell has a cell wall / an animal doesn't; • bacterial cell has no nucleus /an animal cell does; • bacterial cell has a plasmid /the animal cell doesn't; • bacterial cell has a flagellum/ the animal cell doesn't; • bacterial cell has pili /the animal cell doesn't; • bacterial cell has no mitochondria /the animal cell does; • bacterial cell has slime capsule / the animal cell doesn't;		(2)
	(b) (i)	 ; ; ;	Reject more than one line from each part	(3)
	(b) (ii)	• substitution, 5 ÷ 50 000 = 0.0001(cm); • conversion to mm - 0.0001 x 10; • 0.001 mm/1×10^{-3}; or • 5×10=50mm; • 50÷50000; • 0.001mm/1×10^{-3}	ecf for mp 2 and 3 Full marks for correct final answer	(3)

Total for Question 1 = 10 marks

Type MS · Use image size, actual size and magnification

Answer 13 of 13 · Question 4 part b.ii · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

4 (a)		disease causing organism/micro-organism;		1
(b)	(i)	any three from • cytoplasm; • cell/plasma membrane; • cell wall; • (loop of) nucleic acid/DNA; • flagellum;	if more than three parts labelled minus one mark for each incorrect label	3
	(ii)	• distance between A and B = 64 mm/6.4 cm; • 64 × 1000 = 64 000 µm; • magnification = $\frac{64\,000}{2}$ = 32 000;	ecf allow +/- 1 mm	3
(c)	(i)	• suitable scale, axes labelled with units; • correct way round; • points plotted correctly; • lines joining plotted points; • key to identify lines;	needs to cover half of paper award mp1,2 and 5 if bar chart plotted	5
	(ii)	any three from • both fall/vaccine becomes less effective over time; • 45-64 years more effective (over all months)/over 65 less effective (over all months); • slowly for first two months; • more/ rapid /steep decrease, between 3-4 months/after month 3 (for over 65);		3
				15 marks

Type MS · Use image size, actual size and magnification

Answer 13 of 13 · Question 4 part b.ii · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks